Pre-departure Flight Delay Risk Classification

Data Wrangling Group Project: Written Report

Group members: [Member 1] · [Member 2] · [Member 3] | Date: 27 October 2026

Dashboard: public.tableau.com/app/profile/anirudh.rajgopal/viz/FlightDelayInsights_17926536318450

1. Objective

Can we tell, two hours before a flight is scheduled to leave, whether it will depart at least 15 minutes late? We study departures from Atlanta (ATL), Dallas/Fort Worth (DFW) and Chicago O'Hare (ORD). The project takes three public sources (flight records, weather reports and an airport reference table) through extraction, cleaning, transformation and loading. The wrangled data then feeds a binary classifier and an interactive Tableau dashboard.

Main objective: develop and evaluate Logistic Regression, KNN, SVM and Random Forest in a historical simulation with a two-hour prediction cutoff, and judge whether their outputs can identify and prioritise delayed departures. Four supporting objectives structure the evaluation:

- **O1 Model choice:** select a classifier with chronological validation, then confirm it on unseen data.
- **O2 Value of extra data:** measure whether weather, delay history and scheduled traffic (feature set B) improve on schedule-only inputs (feature set A).
- **O3 Failure patterns:** find where the model misses delays or raises false alerts (by airport, airline and time of day).
- **O4 Risk prioritisation:** test how many real delays a 5%, 10% or 20% "watch list" of the riskiest flights captures.

2. Dataset Description

Source	Access method	Content used	Terms / constraints
BTS Reporting Carrier On-Time Performance	Monthly ZIP download (transtats.bts.gov/PREZIP)	Date, airline, flight no., route, scheduled times, distance, duration, cancellation, departure delay	U.S. government public data; no key; large national files (≈0.6M rows/month) read in chunks
Iowa Environmental Mesonet ASOS/METAR	HTTP API request per station and period	Temperature, dew point, wind, gusts, visibility, sky layers, precipitation, raw METAR text; station time zones	Free public archive; no API key; one request per station and period; 'M' marks missing values
OurAirports reference	CSV download	Airport coordinates and elevation (route-feature candidates)	Public-domain data

The data is real, multi-table and large: the twelve 2025 BTS files alone hold 7,001,619 national records, of which 957,315 are scheduled departures at the three hubs. Late-2024 months supply delay history, all of 2025 is used for development, and July 2026 is the untouched test month (87,298 eligible flights). The data is genuinely messy: cancelled flights have no departure outcome, weather fields are frequently unreported (condition codes unknown for 90.6% of training flights, gusts for 80.3%), the weather archive contains duplicate and conflicting reports, visibility is capped at a '10 miles or more' code, and flight times are local HHMM integers that must be converted across time zones before they can be joined to UTC weather. The target, delayed_departure, equals 1 when the departure delay is at least 15 minutes; 25.8% of 2025 training flights and 32.2% of July 2026 test flights are delayed.

3. Methodology by Pipeline Stage

3.1 Extraction

Notebook 01 downloads any missing BTS archive or weather file, stores a SHA-256 checksum for every snapshot and refuses to continue if a re-downloaded file differs from the saved one. National CSVs are read 150,000 rows at a time, and only ATL, DFW and ORD origins are kept. Each flight receives a stable ID made from its source month and original CSV line, so every cleaned row can be traced back to the raw file. Raw files are never modified; acquisition and validation logs are kept in data/raw/_provenance/.

3.2 Cleaning (with log)

Every decision chooses between dropping, imputing or flagging, and the reason is recorded. The table summarises the cleaning log kept in notebook 01 and reports/preparation_manifest.json.

Issue found	Scale	Action	Reason
Cancelled flights (no departure outcome)	18,898 of 957,315 (2025)	Excluded from the target; kept as schedule context for traffic counts	A cancelled flight is not an on-time example; guessing a label would bias the target
Diverted flights	2,750	Kept	Departure delay is still known
Text codes with stray spaces; numbers stored as text	All rows checked	Trimmed; parsed with errors='coerce'; failures counted	0 parse failures, so no values guessed
Duplicate flight schedules	0	Hard stop (assert) if any appear	Never pick a record arbitrarily
Duplicate / conflicting weather reports	4,871 identical rows; 8 conflicting keys	Identical rows collapsed; both conflicting versions excluded	No basis to choose one value
Missing weather values ('M' code)	Gust 80.3%, conditions 90.6% unknown	Left missing + missingness flags; training-median imputation inside the model pipeline	Absent gust is not zero wind; missingness may itself be informative
Trace precipitation ('T')	Present in archive	Numeric 0 plus a separate trace flag	Keeps trace distinct from a true zero
Sky layers missing from structured columns	719 training flights	Re-parsed from raw METAR body	Recovers real ceilings instead of imputing
Distance/duration outliers (IQR rule)	20,787 / 27,641 flagged	Reviewed, not removed or capped	Long routes are valid; deleting them would distort the population
Zero or negative distance/duration	0 found	Rule: set to missing + flag	Physically impossible values should not be trusted
Daylight-saving ambiguous times	Checked	Hard stop if ambiguous	Prevents silent 1-hour errors

Left as-is: very long delays (up to 2,364 minutes) were not capped, because they define the target, and the '10SM' visibility ceiling was kept but documented as '10 miles or more' rather than an exact value.

3.3 Transformation

- **Time alignment:** local HHMM times → minutes after midnight → time-zone-aware timestamps → UTC, so flights and weather share one clock; the prediction cutoff is scheduled departure minus 120 minutes.
- **Merge across tables:** a backward as-of join (pandas merge_asof by origin and time) attaches the latest weather report that is at least 20 minutes older than the cutoff and at most 180 minutes old, so no future weather leaks in.
- **Time-based aggregation:** group-by aggregations over a 28-day window ending 7 days before the cutoff give delay rates for each airport, airline-origin pair, route and flight number, smoothed toward the airport average (prior strength 20) with support counts and no-history flags.
- **Engineered features:** scheduled departures at the origin in the hour before departure (traffic); sine/cosine clock and month coordinates; departure time bands; dew-point spread; altimeter change over 3 hours; gust excess; log and inverse-visibility transforms.

Final feature set A has 26 schedule inputs; feature set B has 66–78 inputs depending on the model (counts before one-hot encoding).

3.4 Loading and Storage

The cleaned flight table (all schedules including cancelled context rows) is written to data/processed/flights_clean.parquet, and the model-ready feature table to features.parquet. Parquet is a typed, compressed, columnar format that preserves dates, nullable integers and categories, unlike a flat CSV. Each load is validated: row counts per source month are logged, checksums of protected files are compared on every rerun (351 files unchanged), and the reproduction check confirmed 3.9 million feature values match the published handoff exactly.

4. Machine Learning Approach and Results

Task: binary classification of delayed_departure. Four algorithms were compared because they learn in different ways: Logistic Regression (a fast, interpretable linear baseline), KNN (similar past flights), RBF SVM (flexible non-linear boundary) and Random Forest (tree ensemble, handles interactions without scaling). They were also compared with four naive baselines: always on time, always delayed, a coin flip, and a historical-delay-rate ranking.

Methodology. A random split would mix future and past flights, so the split is chronological. Settings, feature packages and the decision threshold were chosen with three expanding-window validation folds inside 2025 (train Jan–Mar → score May–Jun; Jan–Jun → Aug–Sep; Jan–Sep → Nov–Dec), selecting by mean delayed-class F1. Final models were refit on a 40,000-flight sample spanning all of 2025 (class proportions preserved) and evaluated once on 10,000 July 2026 flights. Imputation, scaling and one-hot encoding live inside a scikit-learn Pipeline/ColumnTransformer, so they are fitted on training data only.

July 2026 test (B inputs)	F1	Avg. precision	Precision	Recall	ROC-AUC	Accuracy
Logistic Regression (selected)	0.540	0.566	38.1%	92.4%	0.733	49.3%
KNN	0.545	0.532	40.1%	85.3%	0.706	54.2%
RBF SVM	0.534	0.513	43.5%	69.2%	0.681	61.2%
Random Forest	0.543	0.540	39.3%	88.0%	0.723	52.4%
Baseline: always on time	0.000	0.322	–	0.0%	0.500	67.9%
Baseline: always delayed	0.487	0.322	32.2%	100%	0.500	32.2%
Baseline: historical delay rate	0.481	0.396	32.8%	90.4%	0.578	37.3%

Interpretation. Logistic Regression was chosen on validation F1 (0.483) before July was examined; KNN's slightly higher July F1 does not change that choice. All four models beat every baseline on average precision, the threshold-free ranking measure. Accuracy is misleading here: predicting 'on time' for everyone scores 67.9% accuracy yet catches no delays. The honest weakness is precision: the selected threshold catches 92.4% of delays (2,971) but 61.9% of its alerts are false (4,827 false alarms vs. 244 missed delays). The model is good at ranking risk but poor as a yes/no alarm. Logistic Regression is also the cheapest: 0.37 s to train versus about 250 s for SVM.

O2, value of extra data. On identical flights, moving from A to B raised average precision for every model: +6.2 points for LR, +6.7 KNN, +3.1 SVM, +4.8 RF, all with positive 95% day-bootstrap intervals. The combined test cannot separate the contributions of weather, history and traffic.

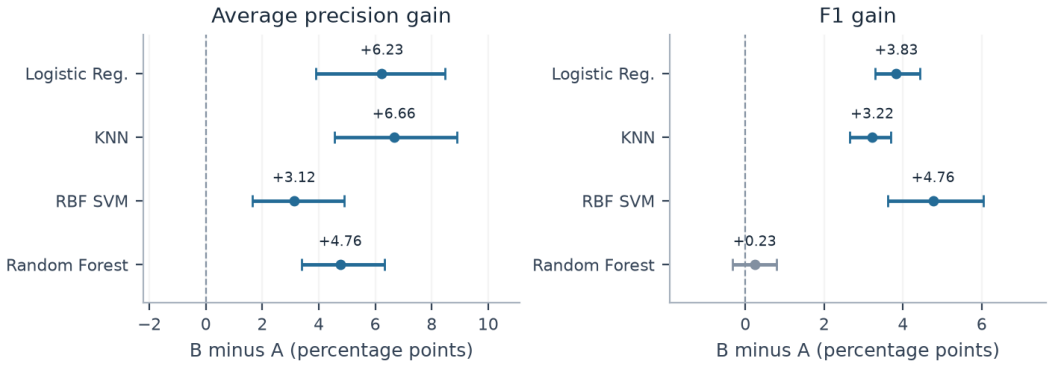


Figure 1. Gain from richer inputs (B minus A) on the same July flights, with 95% day-bootstrap intervals.

O3, failure patterns. False alerts concentrate where delays are common: the evening false-positive rate is 99.7% (the model flags almost every evening flight), ORD's is 75.5% versus 64.7% at ATL, and American Airlines' is 90.6%. Morning flights have the lowest recall (68.5%). Groups below 200 flights or 30 outcomes were not reported.

O4, risk prioritisation. Reviewing only the riskiest 10% (1,000 flights) captured 688 delays (68.8% precision, 2.14× random lift), against 442 for the historical-rate ranking and about 322 at random. The 5% and 20% lists captured 359 and 1,222 delays.

5. Dashboard Walkthrough

The dashboard was built in Tableau Public, connected to the July 2026 test flights with model scores and outcomes exported from notebook 04. It has three pages and is written for a non-technical operations audience: every page has a headline, KPI cards and short plain-English annotations with a suggested action.

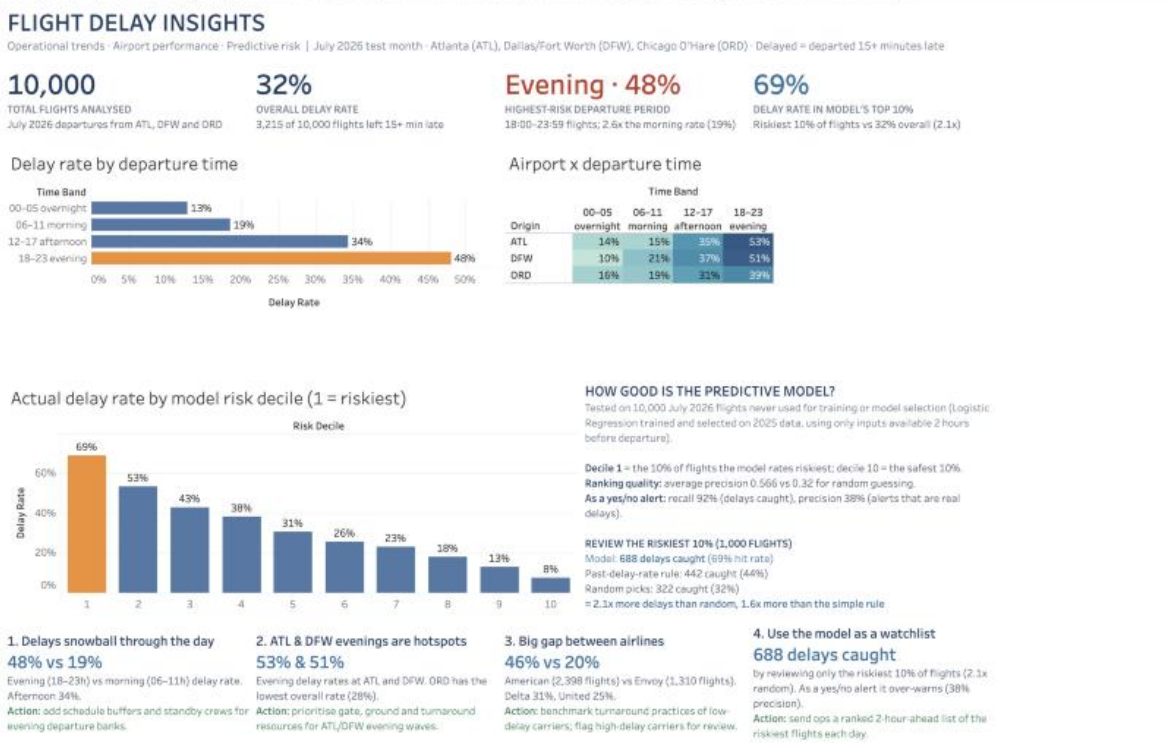


Figure 2. Page 1, Key Insights (overview).

- **Page 1 (Overview):** four KPI cards (10,000 flights, 32% delayed, evening 48% riskiest, 69% delay rate in the model's top 10%); delay rate by time band; an airport × time-band heat map; the risk-decile chart; and four numbered insight cards, each with an 'Action' line.

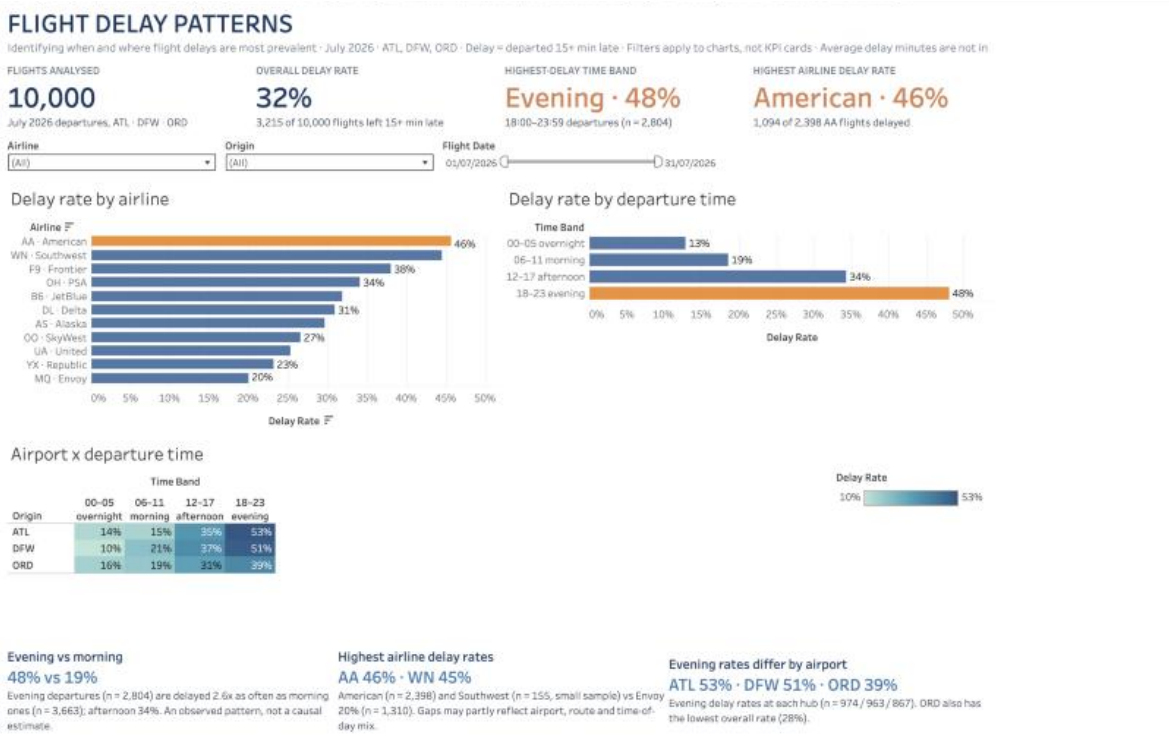


Figure 3. Page 2, Delay Patterns (drill-down with Airline, Origin and Flight Date filters).

- **Page 2 (Detail / drill-down):** interactive Airline and Origin drop-down filters and a Flight Date range slider update the airline ranking, time-band chart and heat map, letting a user drill into one hub, carrier or week. Annotations give sample sizes and warn that the patterns are observational, not causal.



Figure 4. Page 3, Model & Risk Ranking (model insights).

- **Page 3 (Model insights):** headline model metrics (4 models compared, LR selected, AP 0.566 vs. 0.32 random) and the actual delay rate in each model risk decile, falling steadily from 69% in decile 1 to 8% in decile 10, a predicted-versus-actual view showing that the scores rank risk correctly.

6. Key Insights

1. Delays snowball through the day. Delay risk builds through the day: evening departures (18:00–23:59) were delayed 48% of the time versus 19% in the morning, a 2.6× gap. At ATL and DFW the evening rate exceeds 50%, while ORD's is 39%. Knock-on delays from earlier rotations accumulate, so schedule buffers matter most late in the day.

2. A raw airline ranking can reverse once you compare like with like. Frontier (F9) looked worse than American (AA) on raw delay rate (34.6% vs 33.6%, 2025). Once both airlines were given the same mix of airport, month and time band, the order reversed (F9 29.4%, AA 33.4%). Raw airline league tables mostly reflect when and where airlines fly.

3. The model is a better watch list than an alarm. The richest inputs improved ranking (average precision +3 to +7 points) but no model reached 45% precision as a yes/no alarm. The same scores, used as a ranked watch list, find 2.1× more delays than random review and 1.6× more than the simple historical rule. The practical value is in ranking, not in alarms.

4. Accuracy hides failure. A model that always predicts 'on time' is 67.9% accurate but useless. F1, precision-recall and average precision were therefore used for selection and reporting.

7. Limitations

- The test covers one summer month at three hub airports; results may differ in winter, in other years or at smaller airports.
- Excluding cancelled flights means the model predicts delay among flights that operated, not disruption risk in general. On days with mass cancellations (e.g. 643 of 764 ATL schedules on 10 January 2025) the remaining flights are a biased subset.
- The two-hour cutoff is simulated. We assume weather arrives within 20 minutes and past outcomes are available after 7 days; a live feed was not tested.
- Weather fields are heavily missing; median imputation plus flags is a modelling choice that may blur real weather effects. Many flights share one weather report, so they are not independent observations.
- Models were fitted on a 40,000-flight sample (class balance preserved) for practical KNN/SVM run time; rare destinations can be under-represented.
- Scores are not calibrated probabilities, and no cost of a false alarm or missed delay was measured, so we cannot claim money saved or delays prevented. Aircraft rotation was excluded because its timing is unavailable.

Appendix: Reproducibility and Contributions

Run the four notebooks in notebooks/simple/ in order (python3 scripts/run_simple_notebooks.py). Set DOWNLOAD_MISSING = True in notebook 01 to re-download raw BTS, IEM and OurAirports files on a fresh machine. All four notebooks ran end-to-end in fresh kernels with 0 errors (reports/simple_reproduction/).

Member	Primary responsibility
[Member 1]	[e.g. Data extraction, cleaning and cleaning log]
[Member 2]	[e.g. Feature engineering, model training and evaluation]
[Member 3]	[e.g. Tableau dashboard, report and presentation]